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A Data curation

We will list the detail of how we organize and prepare our target and non-target dataset to fine-tune models or evaluating the non-target capabilities.

Programming. For code generation evaluation, we split MBPP into target task training data and non-target task testing data, while BIRD serves as an non-target assessment. This pairing tests both direct programming ability and cross-language generalization from Python to SQL.

- **MBPP:** A Python programming benchmark containing 974 problem-solution pairs. We partition training set into 374 train, 90 validation and using the original 378 test examples. Performance is evaluated using the pass@1 metric.
- **BIRD:** A Text-to-SQL generation benchmark that requires models to translate natural language queries into executable SQL statements. Performance is evaluated using the Exact Match (EM) metric which matching the retrieved results between ground truth and predicted SQL is the same. We use this dataset exclusively for evaluation.

Mathematical reasoning. We split MATH into target task training data and non-target task testing data, and GSM8K for non-target task testing. These datasets differ in format: MATH features competition-style problems, while GSM8K uses natural language, allowing us to assess generalization across mathematical expression styles.